Grandeur: The Consulting & Knowledge cell of SSCBS presents

Industry analysis
Of Pharmaceutical Industry

Industry overview

India is the largest provider of generic drugs globally. The Indian pharmaceutical sector supplies over 50% of global demand for various vaccines, 40% of generic demand in the US and 25% of all medicine in the UK. Globally, India ranks 3rd in terms of pharmaceutical production by volume and 14th by value. The domestic pharmaceutical industry includes a network of 3,000 drug companies and ~10,500 manufacturing units. India enjoys an important position in the global pharmaceuticals sector. The country also has a large pool of scientists and engineers with a potential to steer the industry ahead to greater heights. Presently, over 80% of the antiretroviral drugs used globally to combat AIDS (Acquired Immune Deficiency Syndrome) are supplied by Indian pharmaceutical firms.

SWOT ANALYSIS

STRENGTHS

- Skilled staffs: Any and all staff of the pharmaceutical industry is highly skilled. They are employed only after they have spent a good amount of time on rigorous training.
- Setting the benchmark: Pharmaceutical industry has set an awfully high benchmark for the medical industry. So, by doing its job efficiently the industry is setting a good example of excellence for the medical industry with wide customer base.
- Noticeable market presence: The pharmaceutical industry has quite an impressive presence in the market. Its large and intimidating aura often scares away potential competition and lures in many customers.
- Others: providing quality goods and services and overall excellence. It also includes low operating overhead, firm fiscal management, low staff turnover, high return on investment.

WEAKNESSES

- Non availability of modern technology: This industry lacks modern machinery which makes it rather outdated and dull as customers are always searching for new and improved health care materials. Modern tech supports the production could be a lot faster.
- Lengthy decision-making process: This industry is quite vast and widespread. So, when taking a new and/or revolutionary decision, it may take a while before it is properly executed. As the companies have their own board of directors and chairmans they can hardly agree on anything in a short period of time. This slows down the whole process.

- Low profitability and low profit margins: The price of the resources is quite high in this sector; profits are quite low more often than not. And it's not that it is only the case for some of the companies. Almost all the companies have low profit margins.
- Low staff morale: The staff in this industry doesn't have much motivation. This is the reason for low rates of production. Because many psychological articles suggest that if the morale is low or non-existent the performance of humans takes a heavy blow.

OPPORTUNITIES

- Government support: This industry is one of the industries which is often favoured by the government. So, they get all the support and help they need from the government. It is also the factor for which the industry is flourishing to this day. The laws and taxes are often in their favor.
- Growing space: Financial support: Along with the government, many insurance companies are also favoring this industry. They have reasonable conditions and rules for them to follow. Also, the interest level is very low for them.
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- Patents: The pharmaceutical industry develops new and effective medication. it can patent that product themselves and earn a handsome amount of profit. This gives the industry a chance to grow.
- Global market: Globalization has made the world closer than ever. The pharmaceutical industry is showing a promising future in the global market.

THREATS

- Currency devaluation: Currency devaluation is affecting the pharmaceutical industry in a grim manner. It makes the whole ordeal a lot harder than it already is.
- New competitors: Since globalization, the rivaling industries are increasing. Many people are interested in the Ayurvedic industry, homeopathic industry and many others instead of medicines. This poses a great threat to the pharmaceutical industry.
- Insufficient power supply: The pharmaceutical industry requires a huge amount of power. However, the supply of power on a large scale is a problem. As a result of which many companies of the industry don't get enough power to carry out their necessary routines.
- Inflation: sometimes in some sector is where raw materials are pricey to begin with, inflation of price just adds salt to the wound. Suddenly the necessities are out of reach for many companies in the industry. This causes the production cost to skyrocket and suddenly the profits are lower than ever.

PORTER 5 FORCES

Industry Rivalry

Pharmaceuticals industry is highly competitive because of the potentials for high returns if the new drug is created. The rivalry between the companies is in many forms i.e price competition, advertising battles, product differentiation etc. The rivalry in the industry can be gauged from the fact that the top player in the country has only 6% market share, and the top five players together have about 18% market share. Thus, the concentration ratio for this industry is very low. High growth prospects make it attractive for new players to enter the industry.

Bargaining Power of Buyers

The unique feature of the pharmaceuticals industry is their end-users. Buyers have to buy what the doctor suggests. However, the buyers have the power to influence the price and poses a stronghold over the prices. Buyers could seek price reductions by moving to other suppliers of the product.

Bargaining Power of Suppliers

Suppliers have low bargaining power. Chemicals are required by various pharmaceuticals industry. Chemicals are required for every commodity of pharmaceuticals. If the suppliers are not able to provide them with the chemical on time the companies could shift to the other suppliers without incurring high costs. But sometimes the chemical suppliers can go forward to integrate and form a pharmaceuticals industry. For example-shun chemicals, orchid chemicals, they were chemical suppliers initially but turned themselves into the pharmaceuticals industry.

Threat of Substitutes

This is one of the great advantages of the pharma industry. Whatever happens, demand for pharma products continues and the industry thrives. Pharmaceuticals faces huge competition in terms of substitute goods as the production or creation of generic drugs is cheap. Generic companies pose a threat to non-generic drug manufacturers as they focus on the lower cost of supplies. However, in recent times, the advances made in the field of biotechnology can prove to be a threat to the synthetic pharma industry.

Threat of New Entrants

Pharma industry is one of the most easily accessible industries for an entrepreneur in India. The capital requirement for the industry is very low, creating a regional distribution network is easy, since the point of sales is restricted n India. However, creating brand awareness and franchisee amongst doctors is the key to long-term survival. Also, quality regulations by the government may put some hindrance for establishing new manufacturing operations. Going forward, the impending new patent regime will raise the barriers to entry. But it is unlikely to discourage new entrants, as the market for generics will be as huge.

MAJOR TRENDS

USE OF ARTIFICIAL INTELLIGENCE-The use of AI in the pharmaceutical industry has different roles expected to expand, such as accelerating drug discovery and development. Some of the areas that will benefit from AI include clinical trials, fraud detection, and the overall improvement of medications. AI simplifies the identification of eligibility criteria and the inclusion of patients and also makes the cohort identification process faster and cheaper.

PRECISION MEDICINE-Precision medicine comes from the idea of treating each patient as a unique individual. Advancements in omic and data analysis are providing new insights into how the human body responds to drugs. Although it has posed problems for manufacturers, this is a trend that's expected to continue as methods used are refined. In fact, the number of investors that have embraced precision medicine has been higher than anticipated.

BLOCKCHAIN-As the use of blockchain technology has expanded in the financial arena, others have started to identify ways to use this innovation, including companies in the pharmaceutical industry. The primary purpose of blockchain technology is to simplify the way transactions occur while also optimizing security and transparency without requiring a third party. Blockchain technology is also being explored to tackle the use of counterfeit medicines and substandard drugs that enter into the pharmaceutical supply chain and kill thousands of patients every year.

DIGITAL PROCESSES- The implementation of digital processes isn't new. However, it is something that will gain momentum in the pharmaceutical industry. Many pharmaceutical companies appreciate the implementation of digital processes because the benefits far outweigh the costs. It also improves transparency and has the potential to mitigate the loss of data. The rapid spread of COVID-19 prevents pharma companies from showcasing new products and services. Digital was a weak spot for many pharmaceutical organizations long before 2020. But in 2021, healthcare businesses can't afford to be without a robust online presence.

USE OF CLOUD TECHNOLOGY- Cloud technology enables pharmaceutical companies to partner with multiple stakeholders with a greater level of effectiveness. Implementing a strong infrastructure using cloud technology also offers an opportunity to improve the integrity of data. Just as this technology is expanding across all other industries, it will support the pharmaceutical industry's growth.

COVID Impact

COVID-19, the public health crisis, wasn't expected by anyone. It came and created history, a history that saw many new things happening all across the world. Everything and everyone was affected and hence the Pharmaceutical Industry.
No one in their right mind ever imagined a public health crisis at this scale and hence we weren't able to contain it effectively. COVID-19 impacted everyone, infecting more than 240 million people and taking the lives of approximately 5 million people. But COVID came as a blessing in disguise for the Pharmaceutical Industry, as earlier which was considered as under the carpet industry, was now over the carpet and one of the most important industries in the present times. Before going into why COVID acted as a blessing, let's first look at the challenges that COVID brought for the industry.

Challenges due to COVID

- Supply Chain: The first and foremost challenge for the pharmaceutical industry was Supply Chain. Due to Nationwide lockdowns, transporting important raw materials, and finished medicines was getting difficult. This led to certain problems for the industry.
- Lack of Raw Materials: India imports around 70% of its APIs (Active Pharmaceutical Ingredient) from China. COVID and restriction in foreign trades posed a threat to the nation's security.
- Increased Demand: Due to the pandemic, the demand for certain medicines like Remdesivir, tocilizumab, etc. saw an exponential rise. This was beyond that time production levels which posed the problem for Pharma companies to increase the supply to meet the demand.

Blessing in Disguise

Yes, the Pharmaceutical Industry also saw many challenges and hindrances, but the challenges bought opportunities and benefits with them. We are calling COVID a Blessing in Disguise for Pharmaceutical Industry, and now we would like to explain why we are saying so.

- Limelight: This is the first time in the modern era that a diagnostics-oriented industry has been in the absolute limelight. An industry, which was earlier considered as under the carpet requirement became above the carpet requirement. This bought certain other benefits for the industry like government initiatives and investments, private investments, etc.
- Government Initiatives and Investments: As seen in challenges, India had been dependent on China for most of APIs which is a national health threat. Also, not only India, almost all the world is dependent on China for various inputs like surgical equipment, drug processing, etc. Because of this, the Indian Government took action and created a task force for reviewing the internal API sector. Along with this, India saw many investments in the production of

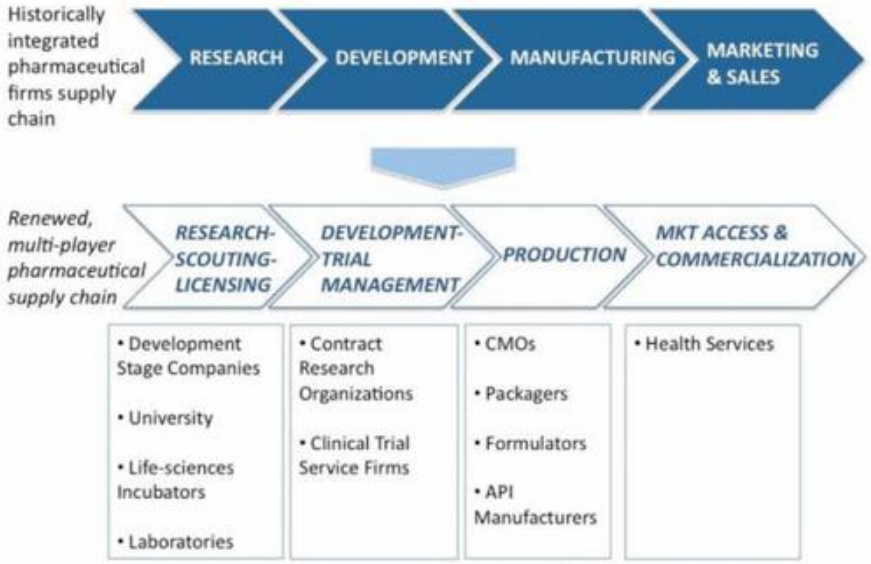
important equipment like masks, ventilators, oxygen generation units, etc. both by private and government.

- Private Investments and Involvement: Due to the lack of many important inputs, private players like corporates, college students, NGOs, etc. took a step in to support the healthcare system of the country. Everyone on their scale supported the industry. This led to multiple small businesses coming into the frame for making masks, college students developing technologies for ventilators, hospitals, oxygen generation plants, etc.
- E-Commerce and Digital Marketing: Earlier a weak spot for the Pharmaceutical Industry, Digital Marketing, and E-Commerce is now a new way for the industry. In the wake of the pandemic, pharma companies have adapted to market themselves digitally and E-pharmacy has taken huge steps in being a solution for the medicinal needs of the people.
- Investments: Earlier we weren't prepared for such a scale pandemic and hence the production capacities were limited, but now, COVID taught us a lesson, and hence, private players, as well as the government, are investing to improve the production capacities of the industry and also making easy for Research and Development of new medicines.

Because of all these developments, we can easily see a bright future for the pharmaceutical industry as now everyone has understood the importance of the same. As the predictions go, Global spending on medicines is expected to reach US$ 1.1 trillion by 2024. Hence, we can safely accept the hypothesis that the COVID-19 pandemic came as a Blessing in Disguise for the Pharmaceutical Industry.

Revenue Model

The world is changing and change is expected in the world. Similarly, the business world is also evolving, and hence comes the change in the business models, supply chains, and revenue models. Traditionally all the businesses were such that most of the tasks were done by them only, but since the mid- 20th century, the term outsourcing is becoming more famous and common. Now businesses are just focusing on one of their main activities and outsourcing the rest to other firms. Outsourcing means the hiring of another firm whose core competency is a particular task for that particular task. This helps in the achievement of specialization.
The pharmaceutical industry is no different from this. To understand the Revenue model of the Pharmaceutical Industry, let's divide it into two parts, traditional and modern, and understand both of them individually.



Stages of the Revenue model of a Pharmaceutical Firm

1. Research: This is the stage where the firms' researched medicines as the cure to various illnesses and diseases. This is one of the most important tasks of a pharmaceutical firm. This is one of the most Investment required stages as Research isn't easy and not all Research reaps benefits. But the one which does get enough benefits for the firm to survive more years
2. Development: This is the stage where the initial phases of the medicine are developed, tested for safety, and getting the patent and other Intellectual Property Rights (IPRs). This is another important stage as the success of the Research depends on the trials. This stage involves investment for making initial medicine, conducting a live trial, and getting government approval and IPRs. This stage ensures that the firm reaps the benefits of its Research
3. Manufacturing: Once the approval is generated for the medicine, the production of the same is started. This is the stage where the firm produces the medicines to

be sold to the customers. This is the phase when the firm makes most of its Working Capital Investment to produce the medicine.

4. Marketing and Sales: Finally comes the stage where the firm realizes the revenues and benefits of its complete investment. This is the stage when the financials of the company show positive inflows and the firm generates enough profits to survive and conduct more research.

In the **traditional model**, all of these tasks are done by the firm itself. This divides the resources of the firms into various tasks, hampering their possible growth. For eg. Let's assume a pharmaceutical company X. X has researched and found a formula to beat disease A. Now X needs to get it approved by the government before starting the production. Let's say it got approval from the government and also some more capital investment from owners due to the success of the Research. Now X has money either to start production or research a new medicine. But X needs to start production to reap the benefits of the current research and hence, X can be late to develop a new medicine.

The **Modern model** caters to this problem. In the modern model, the Research and Development stage is mostly done by the firm, but with the help of other small laboratories and service-providing companies to ease the task of the firm. Production and Marketing tasks are outsourced to other companies, either by selling the license to produce to other players (CMOs, API Manufacturers) or by providing a small margin to them. Both of these ways help the firm save some of its resources which could be used for further research and help the company develop new medicines more quickly than the traditional model.

Nowadays, more companies are adopting the Modern model, because of which, we can see more and more new medicines coming up quickly and medicines quickly reaching most of the population. The best example can be the vaccine for Covid-19, because of licensing of the formula of making the vaccine, we can see many businesses making one kind of vaccine which is allowing everyone to get vaccinated more quickly than ever and hence we can see that by 23 September, approximately 66% of India's adult population received at least the first dose of the vaccine.

Govt Initiatives

The Pharmaceutical Industry is a very important industry in the world. This is the industry which helps in curing the people of various illnesses and hence making them healthy to work again. COVID-19 enhanced the importance of this industry as this industry is now helping in providing vaccines to everyone so that we can go back to pre-COVID time. The government also acknowledges the importance of this industry and hence has taken various initiatives to boost the same. Following are some of the important initiatives taken by the government of India:

- The Indian government is seeking to increase domestic production of bulk drugs and medical devices with a $1.3 billion stimulus plan. (June 17, 2020)
- Under Union Budget 2021-22, the Ministry of Health and Family Welfare has been allocated Rs. 73,932 crore (US$ 10.35 billion) and the Department of Health Research has been allocated Rs. 2,663 crore (US$ 365.68 billion). The government allocated Rs. 37,130 crore (US$ 5.10 billion) to the 'National Health Mission'. PM Atma Nirbhar Swasth Bharat Yojana was allocated Rs. 64,180 crore (US$ 8.80 billion) over six years. The Ministry of AYUSH was allocated Rs. 2,970 crore (US$ 407.84 million), up from Rs. 2,122 crore (US$ 291.39 million).
- The Union Cabinet has given its nod for the amendment of the existing Foreign Direct Investment (FDI) policy in the pharmaceutical sector to allow FDI up to 100% under the automatic route for manufacturing medical devices subject to certain conditions.
- In February 2021, the Punjab government announced the establishment of three pharma parks in the state. Of these, a pharma park has been proposed at Bathinda, spread across ~1,300 acres area and a project worth ~Rs. 1,800 crore (US$ 245.58 million). Another medical park worth Rs. 180 crore (US$ 24.56 million) has been proposed at Rajpura and the third project, a greenfield project, has been proposed at Wazirabad, Fatehgarh Sahib.
- To achieve self-reliance and minimize import dependency in the country's essential bulk drugs, the Department of Pharmaceuticals initiated a PLI scheme to promote domestic manufacturing by setting up greenfield plants with minimum domestic value addition in four separate 'Target Segments' with a cumulative outlay of Rs. 6,940 crore (US$ 951.27 million) from FY21 to FY30.
- In June 2021, Finance Minister Ms. Nirmala Sitharaman announced an additional outlay of Rs. 197,000 crore (the US $26,578.3 million) that will be utilized over five years for the pharmaceutical PLI scheme in 13 key sectors such as active pharmaceutical ingredients, drug intermediaries, and key starting materials.

Investments

The Pharmaceutical industry in India is expected to reach $65 bn by 2024 and to $120 bn by 2030.

The pharmaceutical industry in India is currently valued at $41.7 bn.

Generic drugs, with 71% market share, form the largest segment of the pharmaceutical industry in India.
This is set to grow as exports of generics to the US rise, as branded drugs worth $55 bn will become off-patent during 2017-2019. As per the domestic market share in the pharmaceutical industry by revenue, Anti-Infectives (13.6%), Cardiac (12.4%), and Gastrointestinal (11.5%) had the biggest market share.
Under the Production Linked Incentive (PLI) Scheme for Promotion of Domestic Manufacturing of critical Key Starting Materials (KSMs)/ Drug Intermediates and Active Pharmaceutical Ingredients (APIs), 47 applications with committed investments of INR 5366.35 crore have been approved.
Export of Drugs and Pharmaceuticals was valued at $2001.78 bn in August 2021 with a positive growth of 1.21% over exports of $1977.76 bn in August 2020.

The pharmaceutical industry in India offers 60,000 generic brands across 60 therapeutic categories. The API industry is the third world's largest, and it has 57% of APIs on the WHO.

- Incentives worth INR 21,940 Crore ($3 Mn) are approved for the Indian pharmaceuticals market.
- Expected to reach $65 bn by 2024, and ~$120-130 bn by 2030
- Pharmaceutical industry growth rate 10-12%
- Cost of manufacturing ~ 33% lower than western markets
- 18.7% year on year export growth
- India exports $ 40 K worth of pharmaceuticals every minute

100% Foreign Direct Investment (FDI) in the pharmaceutical sector is allowed under the automatic route for greenfield pharmaceuticals.

100% FDI in the pharmaceutical sector is allowed in brownfield pharmaceuticals; wherein 74% is allowed under the automatic route and thereafter through the government approval route.

Major Investors

COST DRIVERS

1. **MANUFACTURER SELLING PRICE**

Acquisition cost after mfg.

2. **COST, INSURANCE, FREIGHT CHARGES (CIF), IMPORT TARIFFS AND CHARGES**

Cost of importing an Active Pharmaceutical Ingredient (API) or finished product into the country

3. **IMPORTER MARGIN**

Applied by importer who procures and receives the delivering of imported drugs

4. **DISTRIBUTOR MARGIN**

Applied by wholesalers and sub wholesalers for transportation and logistics of storage

5. **RETAILER MARGIN**

Applied by retailers while selling the drugs to the patients (last step of distribution chain)

6. **TAXES**

At both national and regional levels

EMERGING MEDICAL AND HEALTHCARE START-UPS

1. **PharmEasy :** PharmEasy is a company developing a healthcare delivery platform. It helps patients to connect with local pharmacy stores and diagnostic centres enabling them to order medicines, healthcare products, and diagnostic tests. The company aims to improve the supply chain in the pharmaceutical sector by digitizing the process. On September 22, 2020, The Competition Commission of India approved the merger of Medlife (Online Pharmacy) with PharmEasy. This action is termed as the First Major Consolidation in this sector after the entry of Amazon and Reliance. According to this deal, PharmEasy's Parent Entity will acquire 100% equity of Medlife and the promoters of Medlife will get 19.95% stake in the entity

   Buyers - PharmEasy is a ready platform from which the buyers can search for their medicines or healthcare accessories and buy them online without any hassles.

   Suppliers - PharmEasy collaborates with a wide range of local suppliers and medical shops, all of which help the company to arrange their stocks and keep them live online. Besides, the company also earns revenue from various pharmaceutical companies that want to showcase their products online and on the PharmEasy app as featured brands.

   Distribution channel - PharmEasy operates with a vast distribution spread out all across the nation. This helps the company to deliver their products for a broad range of pin codes all over India.

   Revenue Model : PharmEasy primarily earns by displaying the sponsored results of various pharmaceutical entities. These kinds of advertisements are found on the home page of such organizations. Advertising is a major source of revenue and this e-pharmacy leverages it to the hilt. Attractive discounts also contribute to PharmEasy's revenue. Furthermore, PharmEasy earns commission from its customers for the healthcare products and medicines that sell via the platform. The brand also earns through the delivery charges that get levied on the products.

2. **Cure.fit :** Cure.fit is a healthcare start-up that leverages technology and data to enable people to live a healthy lifestyle and get access to healthcare at an affordable cost. The Bengaluru based start-up offers both online and offline experiences across fitness, nutrition and mental wellbeing through its holistically integrated four verticals. While Cult.fit caters to physical fitness, mind.fit is for mental wellbeing, Eat.fit for healthy food, and Care.fit looks at personalised healthcare services, including consultation and diagnostics.

   CureFit was launched in May 2016 by Myntra co-founder Mukesh Bansal, and former Chief Business Officer of Flipkart, Ankit Nagori.. The company has its headquarters located in Bangalore, India. Cure.fit has signed up Hrithik Roshan as its brand ambassador. On acquiring Cult in 2016, the company launched a unique HRX workout with Hrithik Roshan in March 2017, and also acquired

Fitness First, The Tribe, B2B logistics start-up, Opinio, and online food-tech company, Kristy Kitchen. CureFit rebranded two of the three centres of a1000yoga as mind.fit while the third was converted into a Cult fitness centre.

The brand also partnered with Tiger Shroff's clothing line, PROWL in March 2018, for a new fitness program comprising a mix of combat, dance, and functional fitness

In May 2017, Cure.fit launched its mobile app for iOS and Android devices. The app operates on a freemium model. It houses free do-it-yourself (DIY) videos on fitness workouts from home. The app also allows users to book classes, and track progress through a simplified activity points system.

Acquisitions and investments : In July 2016, CureFit raised $15 million from Kalaari, Accel, IDG & UC-RNT. The company invested $3 million and got a majority stake in the Bengaluru-based fitness start-up, Cult in August 2016. In Feb 2017, the company acquired a controlling stake in the fitness centre, The Tribe. In March 2017, Cure.fit launched Eat.fit- a subscription-based food-tech platform by acquiring Opinio and Kristy Kitchen, raised $3 million funding from RNT Capital, Ruchi Sanghvi, Aditya Agarwal and Gokul in June 2017. It also acquired Bengaluru-based Yoga centre - a1000 Yoga in July 2017. In July, Curefit raised $120 million in Series C funding.

KEY PLAYERS

COMPANY	VALUATION
Sun Pharmaceutical Limited	INR 273.28 Billion
Aurobindo Pharma Limited	INR 164.99 Billion
Lupin Limited	INR 159.55 Billion
Cipla Limited	INR 155.77 Billion
Dr. Reddy's Laboratories	INR 144.36 Billion
Cadila Healthcare Limited	INR 120.50 Billion
Intas Pharmaceuticals Limited	INR 108.86 Billion
Glenmark Pharma Limited	INR 91.86 Billion
Torrent Pharmaceuticals Limited	INR 63.01 Billion